

2026-05-25 18:00:07.112584: E external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:467] Unable to register cuFFT factory: Attempting to register factory for plugin cuFFT when one has already been registered

WARNING: All log messages before absl::InitializeLog() is called are written to STDERR

E0000 00:00:1779732007.474242 57 cuda_dnn.cc:8579] Unable to register cuDNN factory: Attempting to register factory for plugin cuDNN when one has already been registered

E0000 00:00:1779732007.605120 57 cuda_blas.cc:1407] Unable to register cuBLAS factory: Attempting to register factory for plugin cuBLAS when one has already been registered

W0000 00:00:1779732008.622914 57 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.

W0000 00:00:1779732008.622960 57 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.

W0000 00:00:1779732008.622965 57 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.

W0000 00:00:1779732008.622968 57 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.

Running on Kaggle

Warning: Looks like you're using an outdated `kaggle` version (installed: 2.0.0), please consider upgrading to the latest version (2.0.2)

Завантаження датасету з Kaggle...

Dataset URL: <https://www.kaggle.com/datasets/tongpython/cat-and-dog>

Завантаження та розпакування завершено.

Створення збалансованої вибірки (по 300 для тренування, по 100 для валідації)...

Датасет-вибірку успішно підготовлено та структуровано!

Ініціалізація генераторів для Inception-v3 (299x299)...

Found 600 images belonging to 2 classes.

Found 200 images belonging to 2 classes.

Ініціалізація генераторів для Mini-Inception (150x150)...

Found 600 images belonging to 2 classes.

Found 200 images belonging to 2 classes.

[ІНФО] Спроба завантаження my_custom_inception_model.keras з GitHub...

[ІНФО] Не вдалося завантажити модель з GitHub (HTTP Error 404: Not Found).

[ІНФО] Починаємо навчання my_custom_inception_model.keras з нуля...

2026-05-25 18:00:52.825812: E external/local_xla/xla/stream_executor/cuda/cuda_platform.cc:51] failed call to cuInit: INTERNAL: CUDA error: Failed call to cuInit: UNKNOWN ERROR (303)

/usr/local/lib/python3.12/dist-packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:121:

UserWarning: Your `PyDataset` class should call `super().\_\_init\_\_(\*\*kwargs)` in its constructor. `\*\*kwargs` can include `workers`, `use\_multiprocessing`, `max\_queue\_size`. Do not pass these arguments to `fit()`, as they will be ignored.

self._warn_if_super_not_called()

Epoch 1/10

18/18  **209s** 11s/step - accuracy: 0.5290 - loss: 0.6992 - val_accuracy: 0.5208 - val_loss: 0.6925

Epoch 2/10

1/18  **2:47** 10s/step - accuracy: 0.6250 - loss: 0.6698

/usr/local/lib/python3.12/dist-packages/keras/src/trainers/epoch_iterator.py:116: UserWarning: Your input ran out of data; interrupting training. Make sure that your dataset or generator can generate at least `steps\_per\_epoch \* epochs` batches. You may need to use the `.repeat()` function when building your dataset.

self._interrupted_warning()

18/18  **24s** 817ms/step - accuracy: 0.6250 - loss: 0.6698 - val_accuracy: 0.5208 - val_loss: 0.6927

Epoch 3/10

18/18  **193s** 11s/step - accuracy: 0.6886 - loss: 0.5985 - val_accuracy: 0.5208 - val_loss: 0.7048

Epoch 4/10

18/18  **41s** 2s/step - accuracy: 0.7500 - loss: 0.5826 - val_accuracy: 0.5208 - val_loss: 0.7048

Epoch 5/10

18/18  **193s** 11s/step - accuracy: 0.6120 - loss: 0.6334 - val_accuracy: 0.5208 - val_loss: 0.7307

Epoch 6/10

18/18  **41s** 2s/step - accuracy: 0.7188 - loss: 0.5679 - val_accuracy: 0.5208 - val_loss: 0.7346

Epoch 7/10

18/18  **192s** 11s/step - accuracy: 0.7023 - loss: 0.5739 - val_accuracy: 0.5208 - val_loss: 0.8203

Epoch 8/10

18/18  **24s** 819ms/step - accuracy: 0.6875 - loss: 0.6100 - val_accuracy: 0.5208 - val_loss: 0.8288

Epoch 9/10

18/18  **192s** 11s/step - accuracy: 0.6753 - loss: 0.6077 - val_accuracy: 0.5208
- val_loss: 0.8628

Epoch 10/10

18/18  **24s** 820ms/step - accuracy: 0.7500 - loss: 0.5393 - val_accuracy:
0.5208 - val_loss: 0.8821

[ІНФО] Збереження навченої моделі у my_custom_inception_model.keras...

[ІНФО] Спроба завантаження mini_inception_model.keras з GitHub...

[ІНФО] Не вдалося завантажити модель з GitHub (HTTP Error 404: Not Found).

[ІНФО] Починаємо навчання mini_inception_model.keras з нуля...

Epoch 1/10

18/18  **9s** 309ms/step - accuracy: 0.5248 - loss: 0.8679 - val_accuracy: 0.4792
- val_loss: 0.7109

Epoch 2/10

18/18  **0s** 21ms/step - accuracy: 0.5312 - loss: 0.8412 - val_accuracy: 0.4844 -
val_loss: 0.7090

Epoch 3/10

18/18  **5s** 282ms/step - accuracy: 0.4783 - loss: 0.8811 - val_accuracy: 0.5729
- val_loss: 0.6908

Epoch 4/10

18/18  **0s** 22ms/step - accuracy: 0.3750 - loss: 1.0028 - val_accuracy: 0.5469 -
val_loss: 0.6904

Epoch 5/10

18/18  **5s** 287ms/step - accuracy: 0.4998 - loss: 0.8282 - val_accuracy: 0.5000
- val_loss: 0.6887

Epoch 6/10

18/18  **0s** 21ms/step - accuracy: 0.5312 - loss: 0.7491 - val_accuracy: 0.5208 -
val_loss: 0.6888

Epoch 7/10

18/18  **10s** 580ms/step - accuracy: 0.5039 - loss: 0.7967 - val_accuracy:
0.5260 - val_loss: 0.6922

Epoch 8/10

18/18  **0s** 21ms/step - accuracy: 0.5938 - loss: 0.6397 - val_accuracy: 0.5260 -
val_loss: 0.6924

Epoch 9/10

18/18 ————— **5s** 284ms/step - accuracy: 0.4914 - loss: 0.7794 - val_accuracy: 0.5208
- val_loss: 0.6963

Epoch 10/10

18/18 ————— **1s** 23ms/step - accuracy: 0.6875 - loss: 0.6345 - val_accuracy: 0.5208 -
val_loss: 0.6966

[ІНФО] Збереження навченої моделі у mini_inception_model.keras...

Генерація прогнозів для валідаційної вибірки (InceptionV3)...

7/7 ————— **16s** 2s/step

--- Метрики якості InceptionV3 ---

	precision	recall	f1-score	support
cats	0.50	1.00	0.67	100
dogs	0.00	0.00	0.00	100
accuracy		0.50		200
macro avg	0.25	0.50	0.33	200
weighted avg	0.25	0.50	0.33	200

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning:
Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero\_division`
parameter to control this behavior.

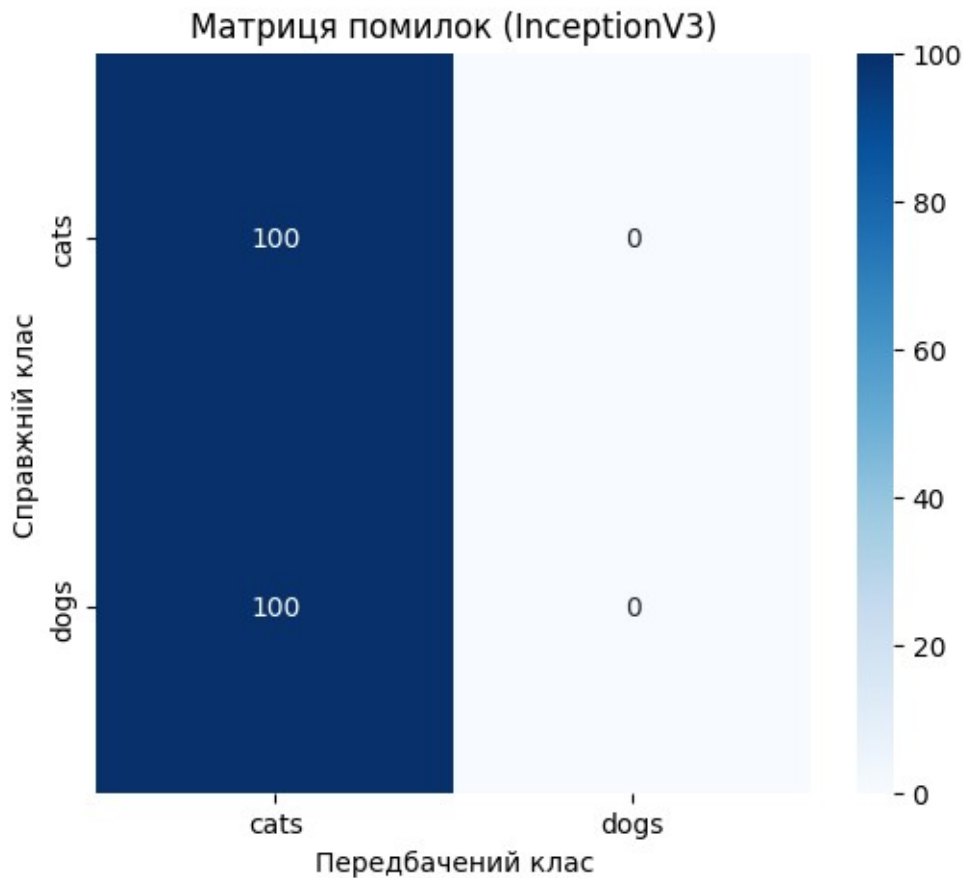
_warn_prf(average, modifier, f'{metric.capitalize()} is', len(result))

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning:
Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero\_division`
parameter to control this behavior.

_warn_prf(average, modifier, f'{metric.capitalize()} is', len(result))

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning:
Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero\_division`
parameter to control this behavior.

_warn_prf(average, modifier, f'{metric.capitalize()} is', len(result))



[ІНФО] Матрицю помилок збережено у: confusion_matrix_InceptionV3.png

Генерація прогнозів для валідаційної вибірки (Mini-Inception)...

7/7 ————— **1s** 63ms/step

--- Метрики якості Mini-Inception ---

	precision	recall	f1-score	support
cats	0.50	1.00	0.67	100
dogs	0.00	0.00	0.00	100
accuracy		0.50		200
macro avg	0.25	0.50	0.33	200
weighted avg	0.25	0.50	0.33	200

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero\_division` parameter to control this behavior.

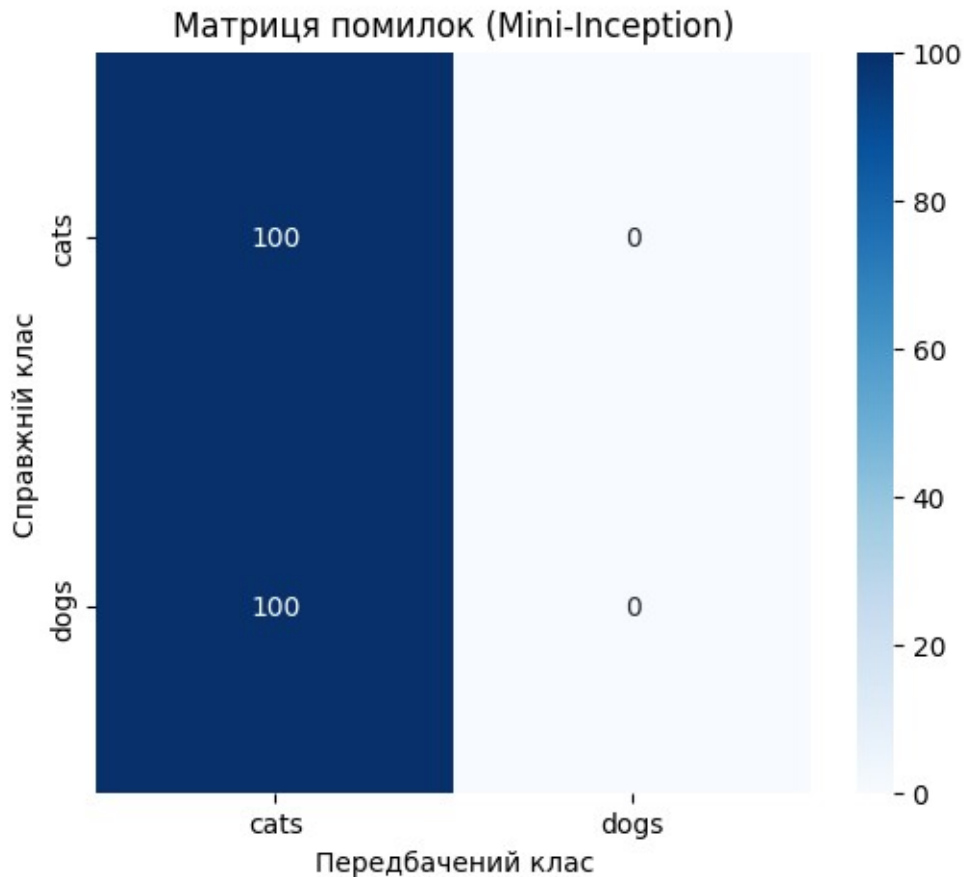
_warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero\_division` parameter to control this behavior.

```
_warn_prf(average, modifier, f'{metric.capitalize()} is', len(result))
```

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero\_division` parameter to control this behavior.

```
_warn_prf(average, modifier, f'{metric.capitalize()} is', len(result))
```



[ІНФО] Матрицю помилок збережено у: confusion_matrix_Mini-Inception.png

=== ДЕМОНСТРАЦІЯ РОБОТИ МОДЕЛЕЙ НА ВИПАДКОВИХ ЗОБРАЖЕННЯХ ===

WARNING:tensorflow:5 out of the last 15 calls to <function

TensorFlowTrainer.make_predict_function.<locals>.one_step_on_data_distributed at 0x7ef8274d76a0>

triggered tf.function retracing. Tracing is expensive and the excessive number of tracings could be due to

(1) creating @tf.function repeatedly in a loop, (2) passing tensors with different shapes, (3) passing Python

objects instead of tensors. For (1), please define your @tf.function outside of the loop. For (2), @tf.function

has reduce_retracing=True option that can avoid unnecessary retracing. For (3), please refer to

https://www.tensorflow.org/guide/function#controlling_retracing and
https://www.tensorflow.org/api_docs/python/tf/function for more details.

WARNING:tensorflow:6 out of the last 16 calls to <function

TensorFlowTrainer.make_predict_function.<locals>.one_step_on_data_distributed at 0x7ef8360572e0>

triggered tf.function retracing. Tracing is expensive and the excessive number of tracings could be due to
 (1) creating @tf.function repeatedly in a loop, (2) passing tensors with different shapes, (3) passing Python
 objects instead of tensors. For (1), please define your @tf.function outside of the loop. For (2), @tf.function
 has reduce_retracing=True option that can avoid unnecessary retracing. For (3), please refer to

https://www.tensorflow.org/guide/function#controlling_retracing and
https://www.tensorflow.org/api_docs/python/tf/function for more details.



[ІНФО] Результати порівняльної демонстрації збережено у: models_demonstration_lab05.png

=== ПОРІВНЯННЯ ОБЧИСЛЮВАЛЬНОЇ СКЛАДНОСТІ ARCHITECTURES ===

1. Специфікація повної кастомної Inception-v3:

Model: "custom_inception_v3"

Layer (type)	Output Shape	Param #	Connected to
--------------	--------------	---------	--------------

input_layer (InputLayer)	(None, 299, 299, 3)	0	-	
conv2d (Conv2D)	(None, 149, 149, 32)	864	input_layer[0][0]	
batch_normalization (BatchNormalizatio...)	(None, 149, 149, 32)	128	conv2d[0][0]	
activation (Activation)	(None, 149, 149, 32)	0	batch_normalizat...	
conv2d_1 (Conv2D)	(None, 147, 147, 32)	9,216	activation[0][0]	
batch_normalizatio... (BatchNormalizatio...)	(None, 147, 147, 32)	128	conv2d_1[0][0]	
activation_1 (Activation)	(None, 147, 147, 32)	0	batch_normalizat...	
conv2d_2 (Conv2D)	(None, 147, 147, 64)	18,432	activation_1[0][...]	
batch_normalizatio... (BatchNormalizatio...)	(None, 147, 147, 64)	256	conv2d_2[0][0]	
activation_2 (Activation)	(None, 147, 147, 64)	0	batch_normalizat...	
max_pooling2d	(None, 73, 73,	0	activation_2[0][...	

(MaxPooling2D)	64		
conv2d_3 (Conv2D)	(None, 73, 73, 80)	5,120	max_pooling2d[0]...
batch_normalizatio...	(None, 73, 73, 80)	320	conv2d_3[0][0]
activation_3	(None, 73, 73, 80)	0	batch_normalizat...
conv2d_4 (Conv2D)	(None, 71, 71, 192)	138,240	activation_3[0][...
batch_normalizatio...	(None, 71, 71, 192)	768	conv2d_4[0][0]
activation_4	(None, 71, 71, 192)	0	batch_normalizat...
max_pooling2d_1	(None, 35, 35, 192)	0	activation_4[0][...
conv2d_8 (Conv2D)	(None, 35, 35, 64)	12,288	max_pooling2d_1[...
batch_normalizatio...	(None, 35, 35, 64)	256	conv2d_8[0][0]
activation_8	(None, 35, 35, 64)	0	batch_normalizat...

conv2d_6 (Conv2D)	(None, 35, 35, 48)	9,216	max_pooling2d_1[...]
conv2d_9 (Conv2D)	(None, 35, 35, 96)	55,296	activation_8[0][...]
batch_normalization_...	(None, 35, 35, 48)	192	conv2d_6[0][0]
batch_normalization_...	(None, 35, 35, 96)	384	conv2d_9[0][0]
activation_6 (Activation)	(None, 35, 35, 48)	0	batch_normalizat...
activation_9 (Activation)	(None, 35, 35, 96)	0	batch_normalizat...
average_pooling2d (AveragePooling2D)	(None, 35, 35, 192)	0	max_pooling2d_1[...]
conv2d_5 (Conv2D)	(None, 35, 35, 64)	12,288	max_pooling2d_1[...]
conv2d_7 (Conv2D)	(None, 35, 35, 64)	27,648	activation_6[0][...]
conv2d_10 (Conv2D)	(None, 35, 35, 96)	82,944	activation_9[0][...]
conv2d_11 (Conv2D)	(None, 35, 35, 32)	6,144	average_pooling2...

batch_normalization_5	(None, 35, 35, 64)	256	conv2d_5[0][0]
batch_normalization_7	(None, 35, 35, 64)	256	conv2d_7[0][0]
batch_normalization_10	(None, 35, 35, 96)	384	conv2d_10[0][0]
batch_normalization_11	(None, 35, 35, 32)	128	conv2d_11[0][0]
activation_5	(None, 35, 35, 64)	0	batch_normalization_11[0][0]
activation_7	(None, 35, 35, 64)	0	batch_normalization_11[0][0]
activation_10	(None, 35, 35, 96)	0	batch_normalization_11[0][0]
activation_11	(None, 35, 35, 32)	0	batch_normalization_11[0][0]
concatenate	(None, 35, 35, 256)	0	activation_5[0][0] + activation_7[0][0] + activation_10[0][0] + activation_11[0][0]
conv2d_13 (Conv2D)	(None, 35, 35, 128)	32,768	concatenate[0][0]

batch_normalization_13	(None, 35, 35, 128)	512	conv2d_13[0][0]
activation_13	(None, 35, 35, 128)	0	batch_normalization_13[0][0]
conv2d_14 (Conv2D)	(None, 35, 35, 128)	114,688	activation_13[0][0]
batch_normalization_14	(None, 35, 35, 128)	512	conv2d_14[0][0]
activation_14	(None, 35, 35, 128)	0	batch_normalization_14[0][0]
average_pooling2d_1	(None, 35, 35, 256)	0	concatenate[0][0]
conv2d_12 (Conv2D)	(None, 35, 35, 192)	49,152	concatenate[0][0]
conv2d_15 (Conv2D)	(None, 35, 35, 192)	172,032	activation_14[0][0]
conv2d_16 (Conv2D)	(None, 35, 35, 192)	49,152	average_pooling2d_1[0][0]
batch_normalization_15	(None, 35, 35, 192)	768	conv2d_12[0][0]
batch_normalization_16	(None, 35, 35, 192)	768	conv2d_15[0][0]

(BatchNormalizatio...	192)		
batch_normalizatio...	(None, 35, 35,	768	conv2d_16[0][0]
(BatchNormalizatio...	192)		
activation_12	(None, 35, 35,	0	batch_normalizat...
(Activation)	192)		
activation_15	(None, 35, 35,	0	batch_normalizat...
(Activation)	192)		
activation_16	(None, 35, 35,	0	batch_normalizat...
(Activation)	192)		
concatenate_1	(None, 35, 35,	0	activation_12[0]...
(Concatenate)	576)		activation_15[0]...
			activation_16[0]...
conv2d_18 (Conv2D)	(None, 35, 35,	221,184	concatenate_1[0]...
	384)		
batch_normalizatio...	(None, 35, 35,	1,536	conv2d_18[0][0]
(BatchNormalizatio...	384)		
activation_18	(None, 35, 35,	0	batch_normalizat...
(Activation)	384)		
conv2d_19 (Conv2D)	(None, 35, 35,	442,368	activation_18[0]...
	384)		
conv2d_20 (Conv2D)	(None, 35, 35,	442,368	activation_18[0]...
	384)		

average_pooling2d_2	(None, 35, 35, 576)	0	concatenate_1[0]...
(AveragePooling2D)			
conv2d_17 (Conv2D)	(None, 35, 35, 320)	184,320	concatenate_1[0]...
batch_normalizatio...	(None, 35, 35, 384)	1,536	conv2d_19[0][0]
(BatchNormalizatio...			
batch_normalizatio...	(None, 35, 35, 384)	1,536	conv2d_20[0][0]
(BatchNormalizatio...			
conv2d_21 (Conv2D)	(None, 35, 35, 192)	110,592	average_pooling2...
batch_normalizatio...	(None, 35, 35, 320)	1,280	conv2d_17[0][0]
(BatchNormalizatio...			
activation_19	(None, 35, 35, 384)	0	batch_normalizati...
(Activation)			
activation_20	(None, 35, 35, 384)	0	batch_normalizati...
(Activation)			
batch_normalizatio...	(None, 35, 35, 192)	768	conv2d_21[0][0]
(BatchNormalizatio...			
activation_17	(None, 35, 35, 320)	0	batch_normalizati...
(Activation)			
concatenate_2	(None, 35, 35, 0)	0	activation_19[0]...

(Concatenate)	768)		activation_20[0]...
activation_21	(None, 35, 35,	0	batch_normalizat...
(Activation)	192)		
concatenate_3	(None, 35, 35,	0	activation_17[0]...
(Concatenate)	1280)		concatenate_2[0]...
			activation_21[0]...
global_average_poo...	(None, 1280)	0	concatenate_3[0]...
(GlobalAveragePool...			
dense (Dense)	(None, 1024)	1,311,744	global_average_p...
dropout (Dropout)	(None, 1024)	0	dense[0][0]
dense_1 (Dense)	(None, 1)	1,025	dropout[0][0]

Total params: 10,554,149 (40.26 MB)

Trainable params: 3,515,809 (13.41 MB)

Non-trainable params: 6,720 (26.25 KB)

Optimizer params: 7,031,620 (26.82 MB)

2. Специфікація оптимізованої Mini-Inception:

Model: "mini_inception"

Layer (type)	Output Shape	Param #	Connected to
input_layer_1 (InputLayer)	(None, 150, 150, 3)	0	-
conv2d_22 (Conv2D)	(None, 75, 75, 16)	432	input_layer_1[0]...

batch_normalizatio...	(None, 75, 75,	64	conv2d_22[0][0]
(BatchNormalizatio...	16)		
activation_22	(None, 75, 75,	0	batch_normalizat...
(Activation)	16)		
max_pooling2d_2	(None, 37, 37,	0	activation_22[0]...
(MaxPooling2D)	16)		
conv2d_24 (Conv2D)	(None, 37, 37,	256	max_pooling2d_2[...
	16)		
batch_normalizatio...	(None, 37, 37,	64	conv2d_24[0][0]
(BatchNormalizatio...	16)		
activation_24	(None, 37, 37,	0	batch_normalizat...
(Activation)	16)		
max_pooling2d_3	(None, 37, 37,	0	max_pooling2d_2[...
(MaxPooling2D)	16)		
conv2d_23 (Conv2D)	(None, 37, 37,	512	max_pooling2d_2[...
	32)		
conv2d_25 (Conv2D)	(None, 37, 37,	4,608	activation_24[0]...
	32)		
conv2d_26 (Conv2D)	(None, 37, 37,	256	max_pooling2d_3[...
	16)		
batch_normalizatio...	(None, 37, 37,	128	conv2d_23[0][0]

(BatchNormalizatio...	32)		
batch_normalizatio...	(None, 37, 37,	128	conv2d_25[0][0]
(BatchNormalizatio...	32)		
batch_normalizatio...	(None, 37, 37,	64	conv2d_26[0][0]
(BatchNormalizatio...	16)		
activation_23	(None, 37, 37,	0	batch_normalizat...
(Activation)	32)		
activation_25	(None, 37, 37,	0	batch_normalizat...
(Activation)	32)		
activation_26	(None, 37, 37,	0	batch_normalizat...
(Activation)	16)		
concatenate_4	(None, 37, 37,	0	activation_23[0]...
(Concatenate)	80)		activation_25[0]...
			activation_26[0]...
global_average_poo...	(None, 80)	0	concatenate_4[0]...
(GlobalAveragePool...			
dense_2 (Dense)	(None, 1)	81	global_average_p...

Total params: 19,333 (75.52 KB)

Trainable params: 6,369 (24.88 KB)

Non-trainable params: 224 (896.00 B)

Optimizer params: 12,740 (49.77 KB)

Висновок: Mini-Inception оперує значно меншою кількістю параметрів, зменшуючи використання оперативної пам'яті (VRAM) та прискорюючи крок навчання, проте

може демонструвати нижчу узагальнюючу здатність на складних вибірках.